MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, SCO/DLA MULTICHIP CMOS

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the detailed requirements for a monolithic CMOS, integrated circuit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2. 1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DODSIS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510		Microcircuits, General Specification For.
MIL-M-55565	-	Microcircuits, Packaging of
MIL-A-48078		Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-883	-	Test Methods and Procedures for Microelectronics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1345

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9333744	Microcircuit, Digital, SCO/DLA Multichip
9355162	Microcircuit, Digital, SCO/DLA Test Specification

2.1.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD and the supplement thereto, if applicable.

(Copies of specifications, standards, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design and Construction. Microcircuit design and construction shall be in accordance with all the requirements specified herein and in the detail specification drawing 9333744.

3.1.1 Physical dimensions. The package dimension shall conform to drawing 9333744.

3.1.2 Package. The finished component shall be a 22 pin hermetically sealed package as shown in drawing 9333744. The lead finish shall be in accordance with MIL-M-38510, Finish A, B, or C.

3.1.3 Lead or terminal material and finish. Lead or terminal material and finish shall conform to the applicable requirements of MIL-M-38510 paragraph 3.5.6.

3.1.4 Pin assignment. The pin assignment shall be in accordance with figure 1.

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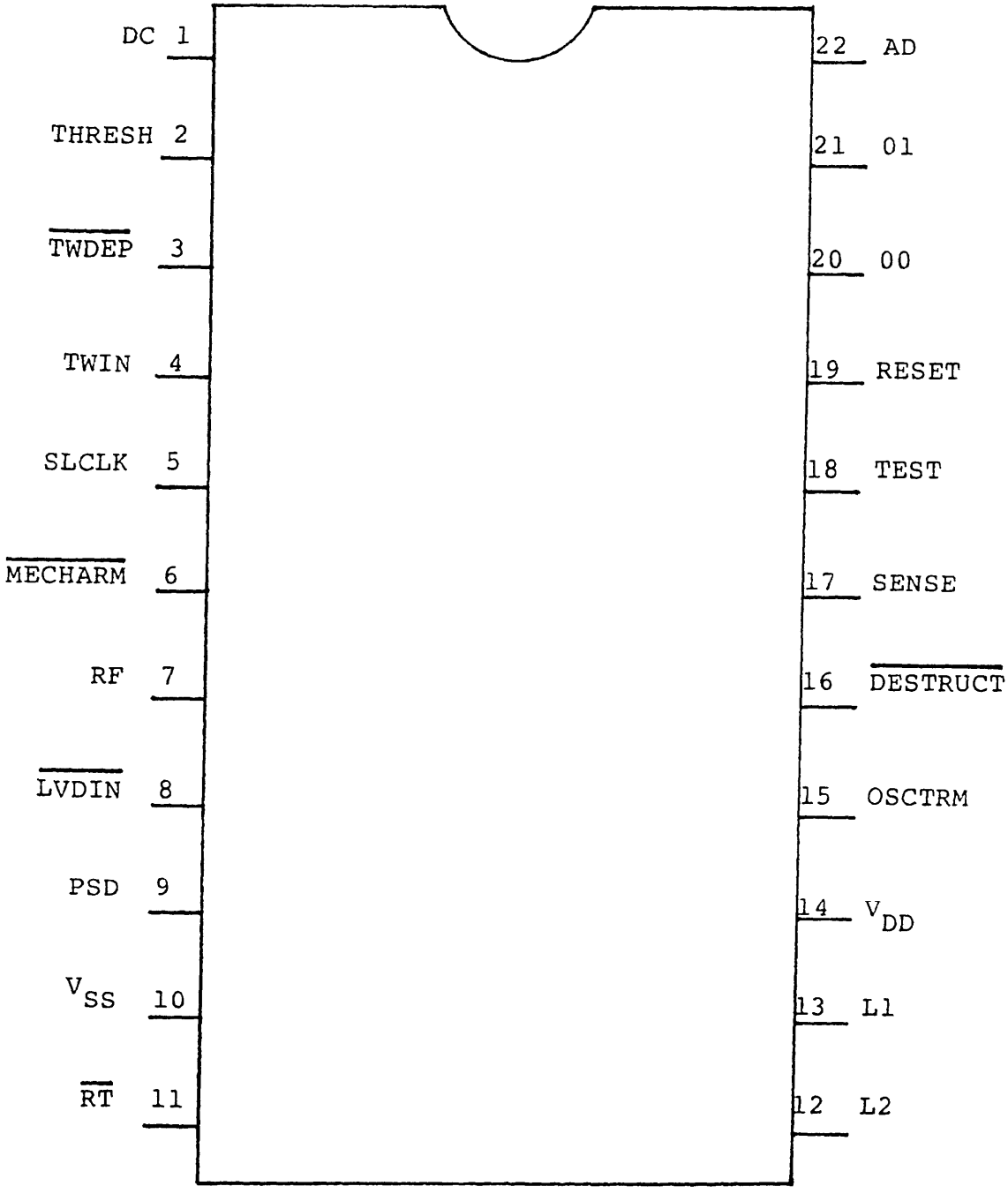


FIGURE 1. PIN ASSIGNMENT

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3.2 Functional description. The functional output of the microcircuit shall be in accordance with all the requirements specified herein and in the test specification 9355162.

3.2.1 General. This component is designed to be used in conjunction with a bipolar R.F. amplifier to provide a miniature, micropower, high frequency? dual conversion superhetrodyne secured data link FM radio reciever. The component shall function primarily as a timer and a special purpose digital pattern decoder. The pattern to be decoded shall hereafter be referred to as the message and shall be in a frequency shift keyed (FSK) Manchester format. The component shall also be electrically programmable. The digital pattern used to program the component shall hereafter be referred to as the field set data. A specified message shall be processed through each of the component electrical circuit functions without operational degradation.

3.3 Environmental and mechanical. Devices to be delivered under this specification shall be capable of meeting the following MIL-STD-883 environmental and mechanical tests when the devices are subjected to these tests in accordance with this specification.

Environmental Tests	Method	Conditions
Moisture resistance	1004	
Steady state life	1005	D, 1000 hrs @ 125°C
Temperature cycling	1010	C, 10 cycles
High temperature storage	1008	C
Seal (Fine and Gross)	1014	(A or B & C) respectively
Burn-in	1015	D, 168 hours @ 125°C

Mechanical Tests	Method	Conditions
Constant acceleration	2001	E, Y1 orientation only
Mechanical shock	2002	B
Solderability	2003	
Lead integrity	2004	B2
Resistance to solvents	2015	
Vibration, variable frequency	2007	A
External visual	2009	
Internal visual	2010	B
Bond strength	2011	D
Physical dimensions	2016	

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3.4 Performance characteristics.

3.4.1 Absolute maximum ratings. The component absolute maximum ratings shall be in accordance with values specified in Table I herein.

TABLE I. Maximum Ratings

CHARACTERISITICS	SYMBOL	RATING	UNITS
Power Supply Voltage	V_{DD}	8.0	V
Input Voltage Range	V_I	-0.5 to V_{DD} +0.5	V
Operating Range 1/	$T_{A(OP)}$	-37° to +57°C	°C
Storage Temp. Range	$T_{A(ST)}$	-55 to +125	°C
Input Current Pins 12 and 13	I_I	±10	A

1/ Component shall be capable of burn-in at +125°C

3.4.2 Operating characteristics. The component shall be operable in the conditions specified in Table II of dwg. 9355162.

3.4.3 Electrical performance characteristics. The component shall meet the electrical performance characteristics specified in Table III of Dwg. 9355162.

3.4.4 Solderability. External leads must be acid tin electroplated or solder dipped per MIL-M-38510, paragraph 3.5.6.2 (a) or (b) and if acid tin electroplated, the minimum plating must be 150 microinches. Parts must also be solderable per MIL-STD-883, method 2003.

3.5 Screening. All devices shall be 100% screened in full accordance with MIL-STD-883, method 5004, class B. Final electrical tests shall include all parameters specified in Tables II and III of Dwg 9355162. Burn-in circuit shall be as shown in figure 2 herein, or equivalent.

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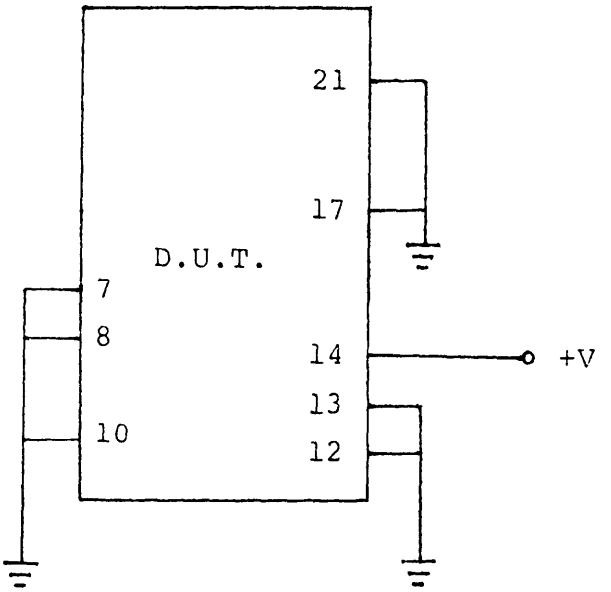


FIGURE 2. BURN-IN CIRCUIT OR EQUIVALENT

3.6 Part marking. Only parts having met all requirements herein shall be marked with the following information in accordance with MIL-M-38510: (A) manufacturer's name or symbol, (B) manufacturer's date code (traceable to the wafer manufacturer's diffusion run), (C) the source control number "1920050CN9333744". Note : Once the part has been marked, the part is classifed confidential and shall be handled as such.

3.6.1 Other marks. The vendor may add additional markings, for his own use, on the bottom surface of the package provided that these markings do not interfere with the required markings.

3.6.2 Resistance to solvent. The marking shall remain legible when subjected to the resistance to solvents testing of MIL-STD-883, Method 2015.

3.7 Process control. All process controls essential for the manufacturing of microcircuits to assure a reliable end item shall be submitted to the designated agency specified in 6.3 herein prior to the start of fabrication.

3.8 Workmanship. (See MIL-M-38510).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions . Unless otherwise specified herein or in the contract, the provisions of MIL-M-38510 shall apply and are hereby made part of this detail specification. The supplier is responsible for the performance of all inspection requirements as specified herein, and the supplier may use his own or any other facilities suitable for the performance of the tests and inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to witness or to perform any of these tests and infections set forth in this specification where such test and inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. The Government reserves the right to audit the data resulting from the manufacturer's performance of these tests and inspections.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection
c. Screening

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2 The first article inspection lot size after screening shall consist of fifty (50) devices, minimum.

4.3.2 Inspection to be performed. First article inspection shall be in accordance with the applicable requirements of MIL-M-38510 and as modified by this specification, excepth that the procuring agency shall be the qualifying activity. First article requirements shall consist of inspections specified in subgroups "at" through "f" listed in Table IV herein. First article items may be subjected by the qualifying activity (Government) to all of the examinations and tests specified in this specification and to any requirements of the applicable drawings.

4.3.3 Rejection. If any device fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Goverment reserves the right to terminate its inspection upon any failure of a device in the sample to comply with any of the stated requirements.

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4.3.4 Requalification. The device shall be subjected to another First Article as stipulated in the contract if:

a. The contractor's manufacturing techniques or process affect the form, fit, function or internal geometry of the device by either the contractor or one of his sub-contractors supplying parts assemblies.
b. The contractor relocated his assembly facilities.
c. The contractor uses a new sub-contractor for material, parts, or assembly.
d. A change occurs in devices material, performance, or assembly.

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TABLE IV. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-M-48646(AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Microcircuit, Digital, SCO/DLA Multichip		1 OF 2	9355162
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
	Screening	50	1/	
	Subgroup a	25		
	Electrical parameters @ -37°C ± 5°C			3.4.3 Table III
	Electrical parameters @ +57°C ± 5°C			3.4.3 Table III
	Physical dimensions			3.3 MIL-STD-883, Method 2016
	Subgroup b	(25)		
	Mechanical shock			3.3 MIL-STD-883, Method 2002, B
	Vibration, variable frequency			3.3 MIL-STD-883, Method 2007, A
	Moisture resistance			3.3 MIL-STD-883, Method 1004
	Seal (fine leak)			3.3 MIL-STD-883, Method 1014, A or B
	Seal (gross leak)			3.3 MIL-STD-883, Method 1014, C
	External visual			3.3,3.8 MIL-STD-883, Method 2009
	Electrical parameters @ 25°C ± 5°C			3.4.3 Table III
	Subgroup c			
	Resistance to solvents	3		3.3 MIL-STD-883, Method 2015
	Bond strength (bond pulls 2/)	(25)		3.3 MIL-STD-883, Method 2011, D

NOTES:

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TABLE IV. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48646(AR)

PARAGRAPH	TITLE: Microcircuit, Digital, SCO/DLA Multichip		SHEET 2 OF 2		DRAWING NUMBER: 9355162
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup d				
	Solderability (leads inspected 3/)	(25)		3.3	MIL-STD-883, Method 2003
	Subgroup e	2			
	Lead integrity			3.3	MIL-STD-883, Method 2004, B
	Seal (fine leak)			3.3	MIL-STD-883, Method 1014, A or B
	Seal (gross leak)			3.3	MIL-STD-883, Method 1014, C
	Subgroup f	25			
	Steady state life			3.3	MIL-STD-883, Method 1005
	Electrical parameters @ 25°C ± 5°C			3.4.3	Table III

NOTES:

1/ Devices which fail any test criteria in the screening sequence shall be removed from the First Article sample and replaced by additional units.

2/ The sample size is the number of bond pulls selected from a minimum number of 10 devices.

3/ The sample size is the number of leads inspected. In no case shall less than 3 devices be used to provide the number of required leads.

4/ All electrical parameters shall be recorded as go/no-go data.

5/ The same sample units shall be subjected to all the tests in a subgroup.

6/ Devices used in subgroup a shall be used in subgroup b. Devices used in subgroups a and b shall be used in subgroups c, d, and e. (Sample 50 units: subgroup a - 25 units; subgroup f - 25 units).

7/ Tests within each subgroup shall be performed in the sequence shown.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots formation shall be as defined in MIL-M-38510 for Class B devices.

4.4.2 Examination.

a. Quality conformance inspection requirements shall be in accordance with Tables I, II, III and specified herein.

b. Inspection requirements. Inspection requirements for this device shall be in accordance with the requirements of MIL-M-38510 and MIL-STD-883B, Method 5005 (Class B).

c. Conformance testing. Groups A, B, C, and D test procedures shall be in accordance with MIL-STD-883, Method 5005, Class B. Group A electrical tests shall include all parameters specified in Tables II and III of Dwg 9355162. Groups C and D end point parameters shall include all parameters specified in Table herein. Sampling plan shall be in accordance with applicable paragraphs of MIL-M-38510, Appendix B.

d. General inspection conditions. Disposition of failed lots and lot resubmission procedures shall be in accordance with applicable paragraphs of MIL-M-38510.

4.5 Test methods, circuits and procedures.

4.5.1 Alternate methods, circuits and procedures. Unless otherwise stated in the specific test method, the methods and circuits shown are given as the basic measurement method. For alternate methods, circuits and procedures the manufacturer shall demonstrate to the procuring activity that alternates which he may desire and propose to use are equivalent and give results within the desired accuracy of measurement (See 6.3).

4.5.2 Test conditions. Unless otherwise specified all examinations and tests shall be made under the following conditions:

Temperature - 25° ± 3°C
Humidity - 10 to 70 percent RH
Pressure - 29 ± 3 inches of Hg

5. PACKAGING

5.1 Packaging. Requirements for packaging shall be in accordance with MIL-M-38510, including provisions for electrostatic protection and MIL-M-55565. Shipping of classified materials shall conform to the requirements established by the Department of Defense for confidential materials using the signature security service.

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6. NOTES

6.1 Intended use. This microcircuit is part of the MOPMS Mine Electronics Assembly.

6.2 Ordering data. Procurement documents shall specifiy the following:

a. Title, number and date of this specification.
b. provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078) Submit equipment designs as required to Commander, US Army Armament Research, Development and Engineering Center, AMSMC-QAT-I(D), Dover, New Jersey 07801-5001.

6.4 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, US Army Armament Research, Development and Engineering Center, AMSMC-QAT-M(D), Dover, New Jersey 07801-5001.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Subject term (key word) listing.

Conformance Testing
Electrical Performance Characteristics
Environmental Testing
First Article Testing
Microcircuit, Digital, SCO/DLA Multichip CMOS
Mechanical Testing
Quality Assurance
Quality Conformance Inspection
Screening

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A344)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-48646 (AR)	Microcircuit, Digital, SCO/DLA Multichip CMOS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark or*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Incl Code)* – Optional
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